The equation of the angle bisectors of (6.37) is

$$\frac{x^2 - y^2}{2} = \frac{xy}{-c}$$

$$\Rightarrow \quad cx^2 + 2xy - cy^2 = 0 \tag{6.39}$$

Given that the angle bisector of (6.37) is (6.38)
Therefore equations

$$x^2 - 2dxy - y^2 = 0$$
$$cx^2 + 2xy - cy^2 = 0$$

represent the same equation as the angle bisector of (6.37)

Comparing the like terms of the above two equations, we get

$$\frac{1}{c} = \frac{-2d}{2} = \frac{-1}{-c}$$
$$\Rightarrow \quad cd = -1$$

### 6.5.4 General form of Pair of Straight Lines

Consider the equations of two arbitrary lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$

The combined equation of the two lines is

$$(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = 0$$

If we multiply the above two factors together, we get a more general equation to a pair of straight lines has the form

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \tag{6.40}$$

The above equation is **a non homogenous equation of degree two.**

in which $a = l_1l_2$ $\quad 2g = l_1n_2 + l_2n_1$
$b = m_1m_2$ $\quad 2f = m_1n_2 + m_2n_1$
$c = n_1n_2$ $\quad 2h = l_1m_2 + l_2m_1$

An equation of the form (6.40) will always represent a pair of straight lines, provided it must able to be factorized into two linear factors of the form $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$.

Condition that the general second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ should represent a pair of straight lines

Let us rearrange the equation of the pair of straight lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ as a quadratic in $x$, we have

$$ax^2 + 2(hy + g)x + \left(by^2 + 2fy + c\right) = 0$$

$$x = \frac{-(hy + g) \pm \sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}}{a}$$

$$ax + hy + g = \pm\sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}$$

That is, $\quad ax + hy + g = \pm\sqrt{(h^2 - ab)y^2 + 2(gh - af)y + g^2 - ac}$